

FLEET RULE OF THE ROAD TEST JANUARY 2015

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(E)Vol I and the Mariners Handbook (NP100).

BR 454 "A Guide to the Collision Avoidance Rules" 6th Edition by Cockcroft and Lameijer was categorised as obsolescent by NCHQ in August 2011. Care should be taken when utilising the 6th Edition as the information contained within may not reflect the most recent changes to the regulations.

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the ROR/training logbook for inspection by FLEET/FOST navigation staff.

In this test, any change in the status of the ship or of the environment will be indicated in the script.

Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90%.

Answers are provided for uniformity of marking. This test is completely scenario based.

You are the NO in HMS DEFENDER (Length 152.4m) alongside in Busan, South Korea. You are due to sail tomorrow to act as Plane Guard for the China Sea US CVN, for a short air defence exercise and are conducting a session of OOW shareholders prior to sailing.

1. The embarked US Officer has been based in Busan for some time and has some local knowledge to impart. He tells the team that the direction of buoyage is reversed in relation to that encountered in the UK.

a. Why is this? (1)

b. Does this change any obligations under the narrow channel rule? (R. 9) (1)

2. There are several local rules outlined for Busan. What do the Rules state about the operation of Special Rules? (R. 1b) (2)

3. One local rule specific to the inland waters of Busan is that a vessel must sound one prolonged blast when getting underway. When un-berthing tomorrow, specifically at what point will you be underway? (R. 3i) **(2)**

After shareholders, you brief the CO on the departure. While examining the charts, he has several questions.

4. The navigation track has been drawn down the centre of the main channel.
- a. How should a vessel navigate down the course of a narrow channel or fairway? (R. 9a) **(1)**
- b. What vessel(s) shall not impede the passage of a vessel which can navigate only within a narrow channel or fairway? (R. 9b) **(1)**
- c. There are several blind bends along the navigation track. What actions should you take when nearing a bend or an area of a narrow channel or fairway where other vessels may be obscured by an intervening obstruction? (R. 9f) **(2)**

The CO approves the departure plan. After a quiet evening alongside you are up on the bridge in the morning to make final preparations for sailing.

5. You sight a vessel in the early morning twilight proceeding outbound in the main channel showing white over red masthead lights in addition to sidelights and sternlight.
- a. What can you deduce about this vessel? (R. 29a) **(2)**
- b. What additional sound signals would this vessel sound in restricted visibility? (R. 35k) **(2)**

The ship closes up SSD and the un-berthing proceeds without incident; the CO hands you the ship and you commence the outbound pilotage.

6. On taking the ship you sight a small power-driven vessel at green 45 at 6 cables crossing the channel. Her bearing is steady. Who is the give-way vessel, what action, if any, are you required to take and what is the obligation of the other vessel given she had already impeded your passage? (R. 9b, 9d, 8f, 15, 34d) **(5)**

The above situation is resolved and the pilotage continues. Looking ahead you sight a group of sailing vessels.

7. How would you tell if these sailing vessels are operating any propelling machinery? (R 25e) **(1)**

8. When 2 sailing vessels are approaching each other so as to involve ROC, with the wind on different sides, who keeps out the way of whom, and how is the windward side defined? (R 12b) **(2)**

The departure continues without incident and you leave the harbour, but remain within a buoyed channel.

9. The OOW reports an outbound merchant vessel closing slowly at red 170 at 2nm.
- a. What situation is this and why? (R. 13b) **(2)**
- b. If you alter course 70° to port in accordance with your navigation track how would this change the situation? (R. 13d) **(2)**

c. Your CO, embarrassed by being overtaken by a merchant vessel, suggests an increase in speed to 20 knots. What do the rules say about safe speed? List the factors that shall be among those taken into account by all vessels. (R 6a) **(4)**

10. You clear the port and fall out SSD. At red 20 at 5nm you sight two small power driven vessels. Their range is closing and both of their bearings are drawing very slowly right. They physically appear to be fishing vessels underway and you confirm this via their AIS transmissions. As you get closer you sight two cones apexes together on both vessels.

a. What do these shapes mean? (R. 26b and c) **(1)**

b. How is this category of vessel defined? (R. 3d) **(1)**

c. What lights would these vessels show by night? (R. 26b) **(3)**

d. What sound signals would this category of vessel make in restricted visibility? (R. 35c) **(2)**

e. In accordance with Rule 18 you make a bold alteration of course to starboard to keep clear of the fishing vessels. Should you use sound signals prior to this manoeuvre? Justify your answer. (R. 34a) **(3)**

You clear these vessels and increase speed to make your ADEX RV. Unfortunately visibility begins to reduce to 1nm.

11. You detect a vessel by radar alone at red 20 at 1.5nm.

a. How is restricted visibility defined in the rules? (R 3l) (1)

b. List your reactions to this contact. (R. 19d) (3)

c. You assess the contact in (a) to be drawing rapidly left with a CPA in excess of 2nm. You now hold a small power driven vessel visually at green 40 at 9c; you are 60° on her port bow and her bearing is drawing very slowly left. Does Rule 19 still apply? Explain how you would manoeuvre for this contact. (R. 19a, 15) (3)

12. How should you use radar equipment, if it is fitted and operational? (R 7b) (2)

Visibility now increases to in excess of 5nm and you continue transit to your RV.

13. You approach the Carrier and her Battle Group and note that all other ships are in station.

a. What does ATP1E state about how ships not in station should manoeuvre? (ATP1E) (1)

b. How should ships in station manoeuvre? (ATP1E) (1)

c. Once in station as plane guard, the USS GEORGE W BUSH commences fixed wing operations. What is the order of priority applicable to Allied naval ships? (ATP1E) (6)

- d. Do the above priorities take precedence over the International Regulations for Preventing Collisions at Sea in this situation? (ATP1E) (1)

The exercise completes at sunset. As you break away you note that the USS GEORGE W BUSH does not have navigation lights in standard positions.

14. What do the Rules say about the operation of special rules with respect to additional station or signal lights? (R. 1c) (2)

Having been detached from the Carrier Group, you begin your transit toward the Malacca Strait and your next port visit.

16. How should a vessel normally leave or join a traffic lane?(R 10b(ii)) (1)

17. What vessels can use an Inshore Traffic Zone if they are not en-route to a port, Or offshore installation? (R10d(i)) (1)

18. What do the rules state about vessels entering a separation zone or crossing a separation line? (R 10E) (2)

You relief appears on the Bridge and commences his handover. You then head down below for dinner.

Total _____ / 65

ANSWERS

You are the NO in HMS DEFENDER (Length 152.4m) alongside in Busan South Korea. You are due to sail tomorrow to act as Plane Guard for the on station US CVN, for a short air defence exercise and are conducting a session of OOW shareholders prior to sailing.

1. The embarked US Officer has been based in Busan previously and has some local knowledge to impart. He tells the team that the direction of buoyage is reversed in relation to that encountered in the UK.

a. Why is this?

Busan, South Korea is IALA Region B vice A (1) or 'red right returning'

b. Does this change any obligations under the narrow channel rule? (R. 9)

No(1)

2. There are several local rules outlined for Busan. What do the rules state about the operation of special rules? (R. 1b)

Nothing in these Rules shall interfere with the operation of special rules made by an appropriate authority for roadsteads, harbours, rivers, lakes, or inland waterways connected with the high seas (1) and navigable by seagoing vessels. Such special rules shall conform as closely as possible to these Rules(1).

3. One local rule specific to inland waters of Busan is that a vessel must sound one prolonged blast when getting underway. When un-berthing tomorrow, specifically at what point will you be underway? (R. 3i)

When the last line is let-go(1) and therefore you are not made fast to the shore(1).

After shareholders, you brief the CO on the departure. While examining the charts, he has several questions.

4. The navigation track has been drawn down the centre of the main channel.

a. How should a vessel navigate down the course of a narrow channel or fairway? (R. 9a)

A vessel proceeding along the course of a narrow channel or fairway shall keep as near to the outer limit of the channel or fairway which lies on her starboard side as is safe and practicable (1).

b. What vessel(s) shall not impede the passage of a vessel which can navigate only within a narrow channel or fairway? (R. 9b)

Vessel of <20m(1/2) or a sailing vessel(1/2).

c. There are several blind bends along the navigation track. What actions should you take when nearing a bend or an area of a narrow channel or fairway where other vessels may be obscured by an intervening obstruction? (R. 9f)

A vessel nearing a bend or an area of narrow channel or fairway where other vessels may be obscured by an intervening obstruction shall navigate with particular alertness and caution(1) and shall sound the appropriate signal prescribed in Rule 34e(1). Namely one prolonged blast.

The CO approves the departure plan. After a quiet evening alongside you are up on the bridge in the morning, to make final preparations for sailing.

5. You sight a vessel in the early morning twilight proceeding outbound in the main channel showing white over red masthead lights in addition to sidelights and sternlight.

- e. What can you deduce about this vessel? (R. 29a)

Pilot vessel on duty (1), underway(1).

- b. What additional sound signals would this vessel sound in restricted visibility? (R. 35k)

Four short blasts(2).

The ship closes up SSD and the un-berthing proceeds without incident; the CO hands you the ship and you commence the outbound pilotage.

6. On taking the ship you sight a small power-driven vessel at green 45 at 6 cables crossing the channel. Her bearing is steady. Who is the give-way vessel, what action, if any, are you required to take and what is the obligation of the other vessel given she had already impeded your passage? (R. 9b, 9d, 8f, 15, 34d) **(5)**

She should not impede your passage. A vessel the passage of which is not to be impeded remains fully obliged to comply with the Rules of this part when two vessels are approaching so as to involve risk of collision (1). You are therefore still required to give way (1). Since you are in a narrow channel, the best course of action is to reduce speed (1). A vessel required not to impede the passage or safe passage of another vessel is not relieved of this obligation if approaching the other vessel so as to involve risk of collision and shall, when taking action, have full regard to the action which may be required by the Rules of this part (2).

The above situation is resolved and the pilotage continues. Looking ahead you sight a group of sailing vessels.

7. How would you tell if these sailing vessels are operating any propelling machinery? (R 25e)

A vessel proceeding under sail when also being propelled by machinery shall exhibit forward where it can best be seen a conical shape, apex downwards. (1)

8. When 2 sailing vessels are approaching each other so as to involve ROC, with the wind on different sides, who keeps out the way of whom, and how is the windward side defined? (R 12b)

When two sailing vessels are approaching one another, so as to involve risk of collision, one of them shall keep out of the way of the other as follows: -

- (i) *When each other has the wind on a different side, the vessel which has the wind on the port side shall keep out of the way of the other;(1)***

For the purpose of this rule the windward side shall be deemed to be the side opposite to that on which the mainsail is carried or, in the case of a square-rigged vessel, the side opposite to that on which the largest fore – and – aft sail is carried.(1)

The departure continues without incident and you leave the harbour, but remain within a buoyed channel.

9. The OOW reports an outbound merchant vessel closing slowly at red 170 at 2nm.

- a. What situation is this and why? (R. 13b)

Overtaking situation (1), the merchant vessel is coming up with your vessel from a direction more than 22.5° abaft your beam (1).

- b. If you alter course 70° to port in accordance with your navigation track how would this change the situation? (R. 13d)

It wouldn't change any obligation under Rule 13 (1), any subsequent alteration of bearing between the two vessels shall not make the overtaking vessel a crossing vessel within the meaning of these rules or relieve her of the duty of keeping clear of the overtaken vessel until she is finally past and clear (1).

c. Your CO, embarrassed by being overtaken by a merchant vessel, suggests an increase in speed to 20 knots. What do the rules say about safe speed? List the factors that shall be among those taken into account by all vessels. (R 6a) **(4)**

Every vessel shall at all times proceed at a safe speed so that she can take proper and effective action to avoid collision and be stopped within a distance appropriate to the prevailing circumstances and conditions (1).

(i) the state of visibility (1/2);

(ii) the traffic density including concentrations of fishing vessels or any other vessels(1/2);

(iii) the manoeuvrability of the vessel with special reference to stopping distance and turning ability in the prevailing conditions(1/2);

(iv) at night the presence of background lights such as from shore lights or from backscatter of her own lights(1/2);

(v) the state of wind, sea and current, and the proximity of navigational hazards(1/2);

(vi) the draught in relation to the available depth of water(1/2).

10. You clear the port and fall out SSD. At red 20 at 5nm you sight two small power driven vessels. Their range is closing and both of their bearings are drawing very slowly right. They physically appear to be fishing vessels underway and you confirm this via their AIS transmissions. As you get closer you sight two cones apexes together on both vessels.

a. What do these shapes mean? (R. 26b and c)

A vessel engaged in fishing/trawling (1).

b. How is this category of vessel defined? (R. 3d)

The term "vessel engaged in fishing" means any vessel fishing with nets, lines, trawls or other fishing apparatus which restrict manoeuvrability, but does not include a vessel fishing with trolling lines or other fishing apparatus which do not restrict manoeuvrability (1).

c. What lights would these vessels show by night? (R. 26b and c)

A vessel engaged in fishing, other than trawling, shall exhibit:

(i) two all-round lights in a vertical line, the upper being red and the lower white, or a shape consisting of two cones with apexes together in a vertical line one above the other;

(ii) when there is outlying gear extending more than 150 metres horizontally from the vessel, an all-round white light or a cone apex upwards in the direction of the gear;

(iii) when making way through the water, in addition to the lights prescribed in this paragraph, sidelights and a sternlight (2).

A vessel when engaged in trawling, by which is meant the dragging through the water of a dredge net or other apparatus used as fishing appliance, shall exhibit:

(i) two all round lights in a vertical line, the upper being green and the lower white, or a shape consisting of two cones apex together in a vertical line one above the other;

(ii) a masthead light abaft of and higher than the all-round green light; a vessel of less than 50m in length shall not be obliged to exhibit such a light but may do so;

(iii) when making way through the water, in addition to the lights prescribed in this paragraph, sidelights and a sternlight.

d. What sound signals would this category of vessel make in restricted visibility? (R. 35c)

One prolonged followed by two short blasts(1) at intervals of not more than 2 minutes(1).

e. In accordance with Rule 18 you make a bold alteration of course to starboard to keep clear of the fishing vessels. Should you use sound signals prior to this manoeuvre? Justify your answer. (R. 34a)

Yes (1), when vessels are in sight of one another, a power-driven vessel underway, when manoeuvring as authorised or required by these Rules, shall indicate that manoeuvre by signals on her whistle (2).

You clear these vessels and increase speed to make your ADEX RV. Unfortunately visibility begins to reduce to 1nm.

11. You detect a vessel by radar alone at red 20 at 1.5nm.

a. How is restricted visibility defined in the rules?

The term restricted visibility means any condition in which visibility is restricted by fog, mist, falling snow, heavy rainstorms, sandstorms or any other similar causes. (1)

b. List your reactions to this contact. (R. 19d) (3)

A vessel which detects by radar alone the presence of another vessel shall determine if a close-quarters situation is developing and/or risk of collision exists (1). If so, she shall take avoiding action in ample time, provided that when such action consists of an alteration in course, so far as possible the following shall be avoided (1):

(i) An alteration of course to port for a vessel forward of the beam, other than for a vessel being overtaken (1/2);

(ii) An alteration of course toward a vessel abeam or abaft the beam (1/2).

c. You assess the contact in (a) to be drawing rapidly left with a CPA in excess of 2nm. You now hold a small power driven vessel visually at green 40 at 9c; you are 60° on her port bow and her bearing is drawing very slowly left. Does Rule 19 still apply? Explain how you would manoeuvre for this contact. (R. 19a, 15) (3)

No. This Rule applies to vessels not in sight of one another when navigating in or near an area of restricted visibility (1). Therefore, Rules in part B section II apply (1). This is therefore a crossing situation and you should manoeuvre in accordance with Rule 15. In this case, an alteration of course to starboard or a reduction in speed would be acceptable (1).

12. How should you use radar equipment, if it is fitted and operational?(R 7b) (2)

Proper use shall be made of radar equipment if fitted and operational, including long range scanning to obtain early warning of risk of collision (1) and radar plotting or equivalent systematic observation of detected objects (1).

Visibility now increases to in excess of 5nm and you continue transit to your RV.

13. You approach the Carrier and her Battle Group and note that all other ships are in station.

- a. What does ATP1E state about how ships not in station should manoeuvre? (ATP1E) (1)

Ships that are not in station are not to hamper those in station.

- b. How should ships in station manoeuvre? (ATP1E) (1)

Ships in station should not stubbornly maintain their course and speed if danger of collision exists.

- c. Once in station as plane guard, the USS GEORGE W BUSH commences fixed wing operations. What is the order of priority applicable to Allied naval ships? (ATP1E) (6)

Priority is in the following order:

(1) Helicopters.

(2) MCM Units.

(3) Ships Engaged in Replenishment.

(4) Ships Engaged in Launching or Recovery of Landing Craft Utilities.

(5) Ships Engaged in the Launching or Recovery of Aircraft.

(6) Ships Engaged in the Launching or Recovery of Arrays.

- d. Do the above priorities take precedence over the International Regulations for Preventing Collisions at Sea in this situation? (ATP1E) (1)

Yes, the priorities above are applicable to Allied naval ships and overrule the International Regulations for Preventing Collisions at Sea in the circumstances described.

The exercise completes at sunset. As you break away you note that the USS GEORGE W BUSH does not have navigation lights in standard positions.

14. What do the Rules say about the operation of special rules with respect to additional station or signal lights? (R. 1c) (2)

Whenever the Government concerned shall have determined that a vessel of special construction or purpose cannot comply fully with the provision of any of these rules with respect to the number, position, range or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signalling appliances, without interfering with the special function of the vessel(1), such vessel shall comply with such other provisions in regard to the number, position, range or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signalling appliances, as her Government shall have determined to be the closest possible compliance with these rules in respect to that vessel(1).

Having been detached from the Carrier Group, you begin your transit toward the Malacca Strait and your next port visit.

15. How should a vessel normally leave or join a traffic lane?(R 10b(ii)) (1)

A vessel using a traffic separation scheme shall:

(iii) normally join or leave a traffic lane at the termination of the lane, but when joining or leaving from either side shall do so at as small an angle to the general direction of traffic as practicable. (1)

16. What vessels can use an Inshore Traffic Zone, if they are not en-route to a port, Or offshore installation? (R 10d(i)) (1)

A vessel shall not use an inshore traffic zone when she can safely use the appropriate traffic lane within the adjacent traffic separation scheme. However, vessels of less than 20m in length, sailing vessels and vessels engaged in fishing may use the inshore traffic zone.(1)

17. What do the rules state about vessels entering a separation zone or crossing a separation line? (R 10e) (2)

A vessel other than a crossing vessel or a vessel joining or leaving a traffic lane shall not normally enter a separation zone or cross a separation line except: -

(i) in cases of emergency to avoid immediate danger(1);

(ii) to engage in fishing within a separation zone(1).

You relief appears on the Bridge and commences his handover. You then head down below for dinner.

Total _____ / 64